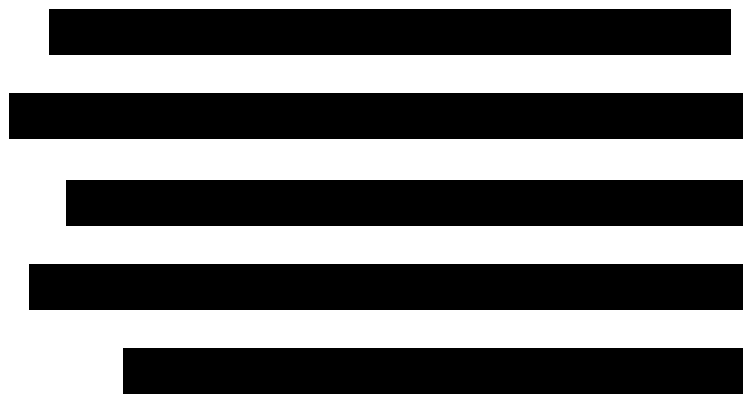


# Group



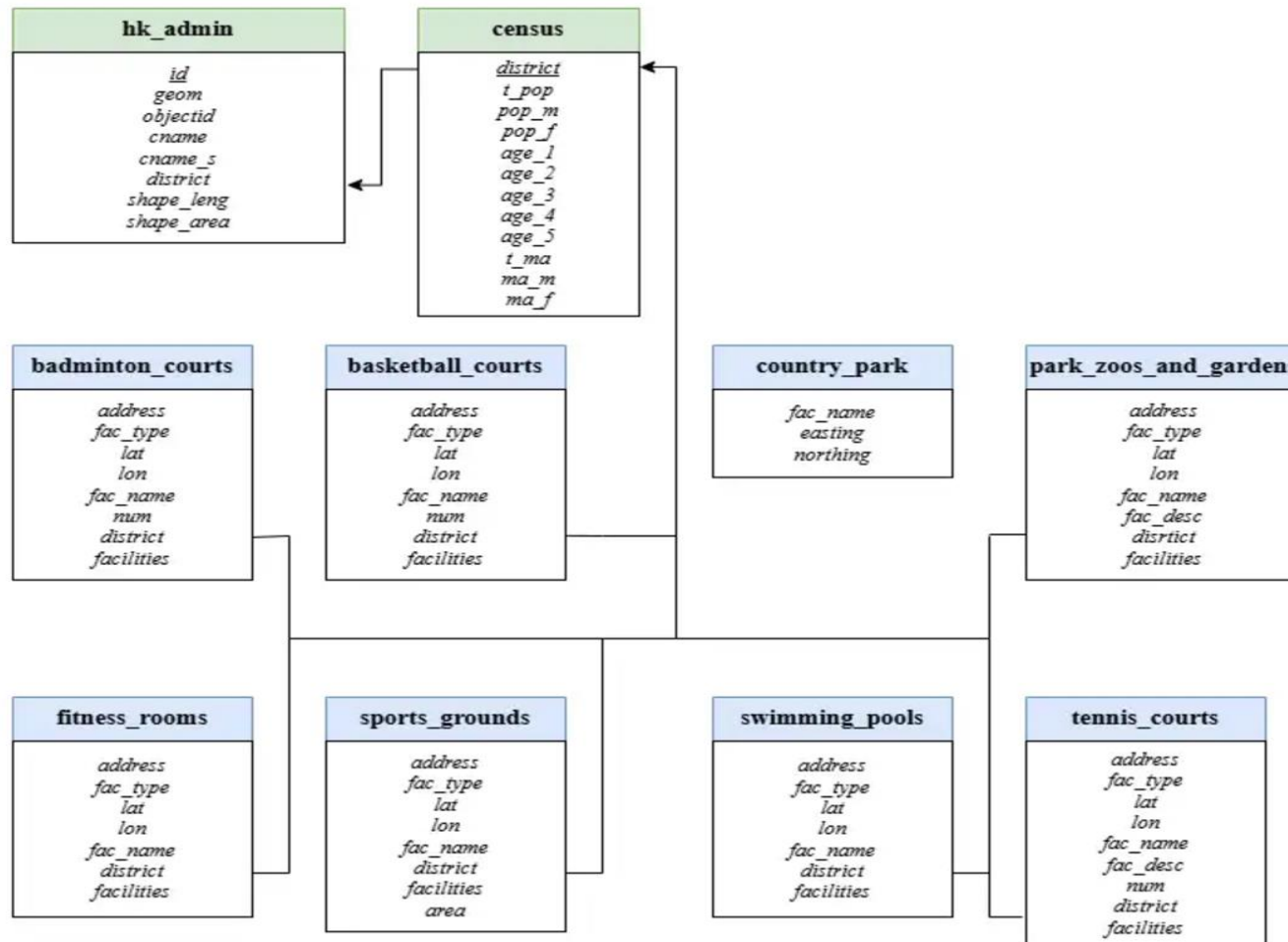
# Data Cleaning(8 different types of facilities)

- ▶ Delete chinese data
- ▶ Delete Nsearch(email address,phone number,etc)
- ▶ Delete opening hour
- ▶ Delete extra geographic location

# Data Cleaning(census)

- ▶ only reserve the population such as age group or sex group

# Database Design



# Initialize tables and helper functions

```
drop table if exists census;
create table census (
  district varchar(30),
  t_pop int,
  pop_m int,
  pop_f int,
  age_1 int, -- <15
  age_2 int, -- 15 - 24
  age_3 int, -- 25 - 44
  age_4 int, -- 45 - 64
  age_5 int, -- 65+
  -- M.A. stands for median age
  t_ma decimal(3, 1),
  ma_m decimal(3, 1),
  ma_f decimal(3, 1),
  primary key(district)
);
```

```
drop table if exists badminton_courts;
create table badminton_courts (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  num int,
  district varchar(30),
  facilities varchar(256)
);
```

```
drop table if exists basketball_courts;
create table basketball_courts (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  num int,
  district varchar(30),
  facilities varchar(256)
);
```

```
drop table if exists country_parks;
create table country_parks (
  fac_name varchar(50),
  easting decimal(10, 4),
  northing decimal(10, 4)
);
```

```
drop table if exists park_zoos_and_gardens;
create table park_zoos_and_gardens (
  address varchar(256),
  fac_type varchar(256),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  fac_desc varchar(384),
  district varchar(30),
  facilities varchar(256)
);
```

```
drop table if exists fitness_rooms;
create table fitness_rooms (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  district varchar(30),
  facilities varchar(256)
);
```

```
drop table if exists sports_grounds;
create table sports_grounds (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(10, 4),
  lon decimal(11, 8),
  fac_name varchar(256),
  district varchar(30),
  facilities varchar(256),
  area decimal(10, 4)
);
```

```
drop table if exists swimming_pools;
create table swimming_pools (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  district varchar(30),
  facilities varchar(256)
);
```

```
drop table if exists tennis_courts;
create table tennis_courts (
  address varchar(256),
  fac_type varchar(30),
  lat decimal(11, 8),
  lon decimal(11, 8),
  fac_name varchar(256),
  fac_desc varchar(256),
  num int,
  district varchar(30),
  facilities varchar(256)
);
```

```
copy census from 'C:\Users\Public\Documents\Postgres16_Group17\census.csv'
delimiter ',' csv header;
```

```
copy badminton_courts from 'C:\Users\Public\Documents\Postgres16_Group17\BadmintonCourts.csv'
delimiter ',' csv header;
```

```
copy basketball_courts from 'C:\Users\Public\Documents\Postgres16_Group17\BasketballCourts.csv'
delimiter ',' csv header;
```

```
copy country_parks from 'C:\Users\Public\Documents\Postgres16_Group17\CountryParks.csv'
delimiter ',' csv header;
```

```
copy fitness_rooms from 'C:\Users\Public\Documents\Postgres16_Group17\FitnessRooms.csv'
delimiter ',' csv header;
```

```
copy park_zoos_and_gardens from 'C:\Users\Public\Documents\Postgres16_Group17\ParkZoosAndGardens.csv'
delimiter ',' csv header;
```

```
copy sports_grounds from 'C:\Users\Public\Documents\Postgres16_Group17\SportsGrounds.csv'
delimiter ',' csv header;
```

```
copy swimming_pools from 'C:\Users\Public\Documents\Postgres16_Group17\SwimmingPools.csv'
delimiter ',' csv header;
```

```
copy tennis_courts from 'C:\Users\Public\Documents\Postgres16_Group17\TennisCourts.csv'
delimiter ',' csv header;
```

```
-- Constructs an HK1980 point, converts to Wgs84
Drop Function If Exists point_hk1980(Numeric, Numeric);
Create Function point_hk1980(easting Numeric, northing Numeric)
Returns Geometry As $$
Begin
  Return ST_Transform(ST_SetSRID(ST_MakePoint(easting, northing), 2326), 4326);
End
$$ Language plpgsql;
```

```
-- Constructs an Wgs84 point
Drop Function If Exists point(Numeric, Numeric);
Create Function point(lat Numeric, lon Numeric)
Returns Geometry As $$
Begin
  -- WGS84 uses (lon, lat) format
  Return ST_SetSRID(ST_MakePoint(lon, lat), 4326);
End;
$$ Language plpgsql;

-- Example
-- ''
-- With geom as (
--   Select geom From hk_admin Where district = 'NORTH'
-- )
-- Select name_en From basketball_courts, geom Where
-- ''

-- Converts meters to degrees
Drop Function If Exists meters(m Float);
Create Function meters(m Float)
Returns Float As $$
Begin
  Return m / 111319.9;
End;
$$ Language plpgsql;
```

# Perform Spatial and Non-Spatial Queries

- ▶ – Given a geographic location (e.g., a lat/lng pair), find the nearest sport facility and its relevant information

```
with nearest as (select fac_type, lat,lon, district,facilities
From Badminton_Courts union
select fac_type, lat,lon, district,facilities
From swimming_pools union
select fac_type, lat,lon, district,facilities
From basketball_courts union
select fac_type, lat,lon, district,facilities
From tennis_courts union
select fac_type, lat,lon, district,facilities
From fitness_rooms union
select fac_type, lat,lon, district,facilities
From sports_grounds)
select fac_type, district,facilities
From nearest
order by ST_Distance(ST_SetSRID(ST_MakePoint(114, 22), 4326),
point(lat, lon))
limit 1
```

|   | fac_type<br>character varying (30) 🔒 | district<br>character varying (30) 🔒 | facilities<br>character varying (256) 🔒 |
|---|--------------------------------------|--------------------------------------|-----------------------------------------|
| 1 | Recreation Grounds                   | ISLANDS                              | TENNIS COURTS                           |

- – Given a geographic location, how many types of sports and outdoor facilities are within a reasonable walking distance?

```
with type_ as (select fac_type, lat,lon, district,facilities
From Badminton_Courts union
select fac_type, lat,lon, district,facilities
From swimming_pools union
select fac_type, lat,lon, district,facilities
From basketball_courts union
select fac_type, lat,lon, district,facilities
From tennis_courts union
select fac_type, lat,lon, district,facilities
From fitness_rooms union
select fac_type, lat,lon, district,facilities
From sports_grounds union
select 'Country Park' as fac_type,
ST_Y(ST_Transform(ST_SetSRID(
ST_MakePoint(easting, northing), 2326), 4326)) as lat,
ST_X(ST_Transform(ST_SetSRID(
ST_MakePoint(easting, northing), 2326), 4326)) as lon,
(select district
from hk_admin, country_parks
where ST_Contains(geom, point_hk1980(easting, northing))
limit 1) as district,
'' as facilities
From country_parks)
select count(distinct fac_type) as number_of_types from type_
where st_dwithin(st_setsrid(st_makepoint(114.181, 22.314), 4236),
st_setsrid(st_makepoint(lon, lat), 4236),
meters(1000)
);
```

|   | number_of_types |                                                                                     |
|---|-----------------|-------------------------------------------------------------------------------------|
|   | bigint          |  |
| 1 |                 | 4                                                                                   |

- For a given type of sports or outdoor facilities, order the administrative districts in Hong Kong by the number of such facilities.

```
select district, count (*) as count_  
From Badminton_Courts  
group by district  
order by count_;
```

|    | district<br>character varying (30) | count_<br>bigint |
|----|------------------------------------|------------------|
| 1  | KOWLOON CITY                       | 4                |
| 2  | SAI KUNG                           | 4                |
| 3  | WAN CHAI                           | 4                |
| 4  | ISLANDS                            | 5                |
| 5  | SOUTHERN                           | 5                |
| 6  | CENTRAL AND WESTERN                | 5                |
| 7  | TSUEN WAN                          | 6                |
| 8  | TAI PO                             | 6                |
| 9  | EASTERN                            | 6                |
| 10 | SHAM SHUI PO                       | 6                |
| 11 | WONG TAI SIN                       | 7                |
| 12 | TUEN MUN                           | 7                |
| 13 | YAU TSIM MONG                      | 7                |
| 14 | SHA TIN                            | 7                |
| 15 | YUEN LONG                          | 8                |
| 16 | KWAI TSING                         | 9                |
| 17 | NORTH                              | 10               |
| 18 | KWUN TONG                          | 10               |

- – For a given type of sports or outdoor facilities, order the administrative districts in Hong Kong by the per capita of facility count.

```
with types_ as (select district, count(*) as count_  
From Badminton_Courts b  
group by district)  
select distinct t.district, c.t_pop::float/count_ as per_capita  
From types_ t, census c  
where t.district=c.district  
order by per_capita;
```

|    | district<br>character varying (30) | per_capita<br>double precision |
|----|------------------------------------|--------------------------------|
| 1  | NORTH                              | 30963.1                        |
| 2  | ISLANDS                            | 37056.4                        |
| 3  | WAN CHAI                           | 41673.75                       |
| 4  | YAU TSIM MONG                      | 44378.142857142855             |
| 5  | CENTRAL AND WESTERN                | 47190.6                        |
| 6  | SOUTHERN                           | 52655.6                        |
| 7  | TAI PO                             | 52745                          |
| 8  | TSUEN WAN                          | 53349                          |
| 9  | KWAI TSING                         | 55088.666666666664             |
| 10 | WONG TAI SIN                       | 58114.57142857143              |
| 11 | KWUN TONG                          | 67316.6                        |
| 12 | SHAM SHUI PO                       | 71848.33333333333              |
| 13 | TUEN MUN                           | 72411.28571428571              |
| 14 | YUEN LONG                          | 83510                          |
| 15 | EASTERN                            | 88267.16666666667              |
| 16 | SHA TIN                            | 98972.28571428571              |
| 17 | KOWLOON CITY                       | 102658.5                       |
| 18 | SAI KUNG                           | 122259.25                      |

# Find the number of fitness room and male ratio in different districts

```
1 with i as (select district, count(district) as number_fitness_room
2 from fitness_rooms f
3 group by district)
4 select c.district, c.pop_m::float/c.t_pop::float *100 as percentage, i.number_fitness_room
5 from census c, i
6 where c.district=i.district
7 order by percentage desc
```



# Find the number of fitness room and male ratio in different districts

| Data Output Messages Notifications |                                         |                                |                               |
|------------------------------------|-----------------------------------------|--------------------------------|-------------------------------|
|                                    | district<br>[PK] character varying (30) | percentage<br>double precision | number_fitness_room<br>bigint |
| 1                                  | TUEN MUN                                | 47.411906983718005             | 5                             |
| 2                                  | YUEN LONG                               | 47.008741468087656             | 6                             |
| 3                                  | NORTH                                   | 46.9407133006708               | 5                             |
| 4                                  | KWAI TSING                              | 46.83621152162776              | 7                             |
| 5                                  | KWUN TONG                               | 46.39316305339248              | 5                             |
| 6                                  | TAI PO                                  | 46.14592220431636              | 5                             |
| 7                                  | WONG TAI SIN                            | 46.098839238745136             | 3                             |
| 8                                  | SHAM SHUI PO                            | 45.90178385024009              | 3                             |
| 9                                  | SHA TIN                                 | 45.88975846052142              | 5                             |
| 10                                 | ISLANDS                                 | 45.867920251292624             | 4                             |
| 11                                 | TSUEN WAN                               | 45.7403137828263               | 4                             |
| 12                                 | SAI KUNG                                | 45.61904313988512              | 4                             |
| 13                                 | YAU TSIM MONG                           | 44.7044394441279               | 4                             |
| 14                                 | SOUTHERN                                | 44.15066963437887              | 2                             |
| 15                                 | EASTERN                                 | 44.09755987031795              | 3                             |
| 16                                 | KOWLOON CITY                            | 43.717276211906466             | 3                             |
| 17                                 | CENTRAL AND WESTERN                     | 41.653210597025684             | 5                             |
| 18                                 | WAN CHAI                                | 40.23815951288281              | 1                             |

# Find the distance between the sports grounds and park with zoos and gardens in the most populated district by age >65

```
1 with p as(select district
2   from census
3   order by age_5 desc
4   limit 1 ),l as (
5   select *
6   from park_zoos_and_gardens z
7   where z.district = (SELECT district FROM p)), k as(
8   select *
9   from sports_grounds
10  where district=(SELECT district FROM p))
11  select l.fac_name,k.fac_name,st_distance (point(l.lat,l.lon)::geography,point(k.lat,k.lon)::geography) as distance
12  from p, l,k
13  order by distance
```



Find the distance between the sports grounds and park with zoos and gardens in the most populated district by age >65

| Data Output                                                                       |                                                                                    | Messages                                                                          | Notifications                                                                     |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|  |   |  |  |
|  |   |  |  |
|  |  |                                                                                   |                                                                                   |

|   | fac_name<br>character varying (256)   | fac_name<br>character varying (256) | distance<br>double precision |
|---|---------------------------------------|-------------------------------------|------------------------------|
| 1 | Ngau Tau Kok Park                     | Kowloon Bay Sports Ground           | 695.74117083                 |
| 2 | Jordan Valley Park                    | Kowloon Bay Sports Ground           | 1274.15937649                |
| 3 | On Sau Road Park                      | Kowloon Bay Sports Ground           | 2331.28269033                |
| 4 | Wai Lok Street Temporary Soccer Pitch | Kowloon Bay Sports Ground           | 3048.59356708                |



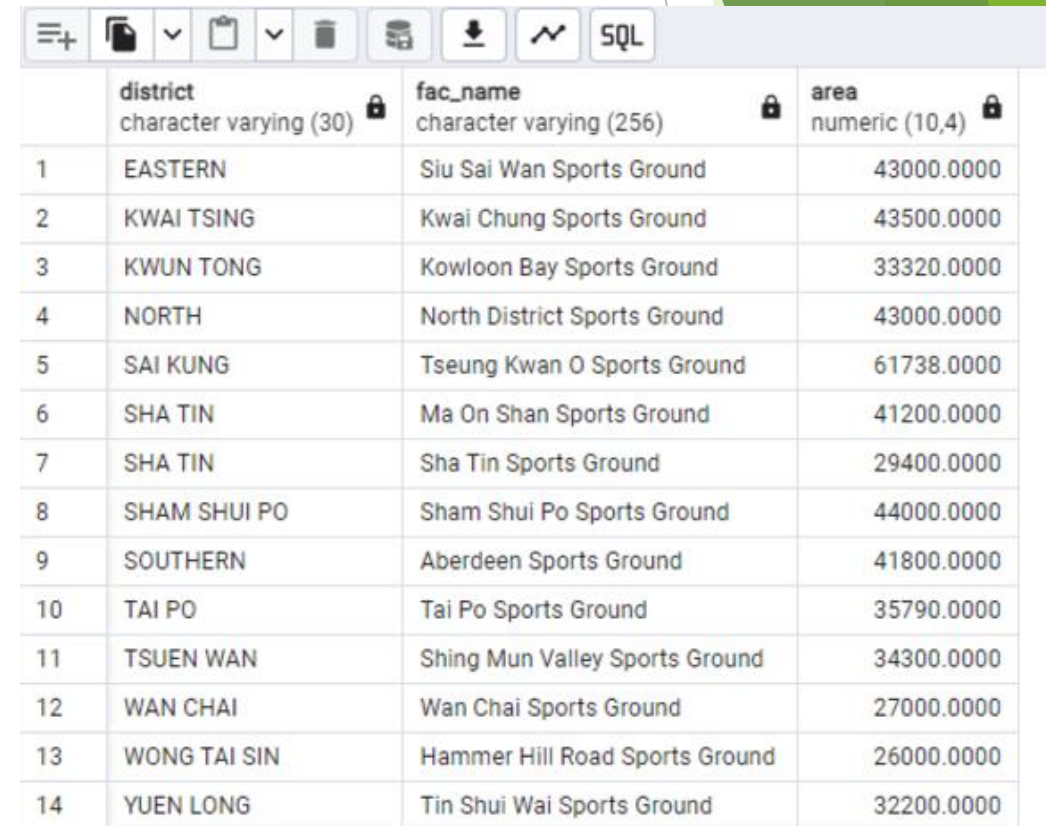
# Find the 3 country parks closest to the centroid of Kowloon City district

```
1  -- Identify the 3 country parks that are closest to the
2  -- geometric center (centroid) of the 'KOWLOON CITY' district
3  WITH kowloon_city_geom AS (
4      SELECT ST_Centroid(geom) AS centroid
5      FROM hk_admin
6      WHERE district = 'KOWLOON CITY'
7  ), country_parks AS (
8      SELECT fac_name, point_hk1980(easting, northing) AS point
9      FROM country_parks
10 )
11 SELECT cp.fac_name, ST_Distance(kcg.centroid, cp.point) AS distance
12 FROM country_parks cp
13 CROSS JOIN kowloon_city_geom kcg
14 ORDER BY distance
15 LIMIT 3;
```

| Data Output |                                        |                              | Messages | Notifications |
|-------------|----------------------------------------|------------------------------|----------|---------------|
|             |                                        |                              | SQL      |               |
|             | fac_name<br>character varying (50)     | distance<br>double precision |          |               |
| 1           | Lion Rock Country Park                 | 0.03703648873766264          |          |               |
| 2           | Kam Shan Country Park                  | 0.05452523226542219          |          |               |
| 3           | Tai Tam Country Park (Quarry Bay Ext.) | 0.05695022843875633          |          |               |

# Find all sports grounds that are top 2 in their districts, and with area greater than 25000 sqm

```
1  -- Select all sports grounds 1st or 2nd largest in
2  -- their districts and areas greater than 25000 sqm
3
4  WITH district_sports_grounds AS (
5      SELECT
6          sg.district,
7          sg.fac_name,
8          sg.area,
9          -- rank of sports ground area within district
10         ROW_NUMBER() OVER (PARTITION BY sg.district ORDER BY sg.area DESC) AS rn
11     FROM sports_grounds sg
12 )
13 SELECT dsg.district, dsg.fac_name, dsg.area
14 FROM district_sports_grounds dsg
15 WHERE dsg.rn <= 2 AND dsg.area > 25000;
```



|    | district<br>character varying (30) | fac_name<br>character varying (256) | area<br>numeric (10,4) |
|----|------------------------------------|-------------------------------------|------------------------|
| 1  | EASTERN                            | Siu Sai Wan Sports Ground           | 43000.0000             |
| 2  | KWAI TSING                         | Kwai Chung Sports Ground            | 43500.0000             |
| 3  | KWUN TONG                          | Kowloon Bay Sports Ground           | 33320.0000             |
| 4  | NORTH                              | North District Sports Ground        | 43000.0000             |
| 5  | SAI KUNG                           | Tseung Kwan O Sports Ground         | 61738.0000             |
| 6  | SHA TIN                            | Ma On Shan Sports Ground            | 41200.0000             |
| 7  | SHA TIN                            | Sha Tin Sports Ground               | 29400.0000             |
| 8  | SHAM SHUI PO                       | Sham Shui Po Sports Ground          | 44000.0000             |
| 9  | SOUTHERN                           | Aberdeen Sports Ground              | 41800.0000             |
| 10 | TAI PO                             | Tai Po Sports Ground                | 35790.0000             |
| 11 | TSUEN WAN                          | Shing Mun Valley Sports Ground      | 34300.0000             |
| 12 | WAN CHAI                           | Wan Chai Sports Ground              | 27000.0000             |
| 13 | WONG TAI SIN                       | Hammer Hill Road Sports Ground      | 26000.0000             |
| 14 | YUEN LONG                          | Tin Shui Wai Sports Ground          | 32200.0000             |

# How many people aged >65 share one park or sport ground in highest population of age >65

```
WITH age as (SELECT district FROM census
              order by age_5 DESC
              limit 1),
park as (
  (SELECT p.district FROM public.sports_grounds p,age a
   WHERE p.district = a.district)
  UNION ALL
  (SELECT p.district FROM public.park_zoos_and_gardens p,age a
   WHERE p.district = a.district)
),
num as (
  SELECT count(*) as count_num
  FROM park
)
SELECT a.district,(c.age_5::float/n.count_num) as average
FROM census c, num n, age a
WHERE c.district = a.district;
```

|   | district<br>[PK] character varying (30)  | average<br>double precision  |
|---|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1 | KWUN TONG                                                                                                                     | 29548.2                                                                                                           |

# Find out the top 5 closest fitness center from the least number of fitness center district

```
WITH mid as (  
    SELECT count(district) as num, district  
    FROM public.fitness_rooms  
    group by district  
    ORDER BY num asc  
    LIMIT 1  
)  
,  
fitness as (  
    SELECT *  
    FROM public.fitness_rooms f  
    WHERE f.district = (SELECT m.district FROM mid m)  
    Limit 1  
)  
SELECT f.fac_name,pf.fac_name, st_distance (point(f.lat,f.lon)::geography,point(pf.lat,pf.lon)::geography) as distance  
FROM fitness f,public.fitness_rooms pf  
order by distance  
limit 6;
```

|   | fac_name<br>character varying (256) | fac_name<br>character varying (256)     | distance<br>double precision |
|---|-------------------------------------|-----------------------------------------|------------------------------|
| 1 | Lockhart Road Sports Centre         | Lockhart Road Sports Centre             | 0                            |
| 2 | Lockhart Road Sports Centre         | Hong Kong Park Sports Centre            | 1663.97305365                |
| 3 | Lockhart Road Sports Centre         | Kowloon Park Sports Centre              | 2690.15159652                |
| 4 | Lockhart Road Sports Centre         | Sheung Wan Sports Centre                | 2799.2551512                 |
| 5 | Lockhart Road Sports Centre         | Kwun Chung Sports Centre                | 3010.84504515                |
| 6 | Lockhart Road Sports Centre         | Sun Yat Sen Memorial Park Sports Centre | 3552.41767856                |

- Find out how far is the SportsGrounds in the largest area from Tai Po District

```
1  ✓ With I as (Select point(lat,lon)as point
2    From sports_grounds
3    order by area desc
4    limit 1),district as
5    (Select geom
6    from hk_admin
7    where district='TAI PO')
8    Select ST_distance(I.point::geography,district.geom::geography)as distance
9    from I, district;
```

|   | distance<br>double precision  |
|---|------------------------------------------------------------------------------------------------------------------|
| 1 | 9097.29469831                                                                                                    |

In this way we know the distance from Tai Po District to the largest sportsground in Hong Kong.

**Distance=9097m**

## ➤ Find all swimming pool addresses in the most populous area

```
1 with we as(select district,max(t_pop) as a
2 from census
3 group by district
4 order by a desc
5 limit 1)
6 select we.district, s.address, c.t_pop
7 from we,swimming_pools s, census c
8 where we.district=s.district and c.district=s.district
9
```

|   | district<br>character varying (30) | address<br>character varying (256) | t_pop<br>integer |
|---|------------------------------------|------------------------------------|------------------|
| 1 | SHA TIN                            | 10 Yuen Wo Road, Sha Tin           | 692806           |
| 2 | SHA TIN                            | 33 On Chun Street, Ma On Shan      | 692806           |
| 3 | SHA TIN                            | 68 Che Kung Miu Road, Tai Wai      | 692806           |

Since it has the largest population, it avoids overcrowding when swimming. By finding the addresses of 3 swimming pools, we let people know where the swimming pools are so that they can spread out the crowd and let people have more fun.

- The population density of people under the age of 15 using parks in each district of Hong Kong

```
1 SELECT p.district, (c.age_1/COUNT(*)) as a
2 FROM park_zoos_and_gardens p, census c
3 WHERE p.district = c.district
4 GROUP BY p.district, c.age_1
5 ORDER BY a;
```

| Data Output Messages Notifications |                                    |             |
|------------------------------------|------------------------------------|-------------|
| ≡+ 📄 ▼ 🗑️ ⚙️ ⬇️ ⌵ SQL              |                                    |             |
|                                    | district<br>character varying (30) | a<br>bigint |
| 1                                  | WAN CHAI                           | 2114        |
| 2                                  | CENTRAL AND WESTERN                | 2438        |
| 3                                  | SOUTHERN                           | 4542        |
| 4                                  | ISLANDS                            | 4753        |
| 5                                  | TAI PO                             | 4911        |
| 6                                  | WONG TAI SIN                       | 5181        |
| 7                                  | SAI KUNG                           | 6258        |
| 8                                  | NORTH                              | 6699        |
| 9                                  | TUEN MUN                           | 6767        |
| 10                                 | SHAM SHUI PO                       | 7048        |
| 11                                 | KWAI TSING                         | 8236        |
| 12                                 | SHA TIN                            | 8765        |
| 13                                 | YAU TSIM MONG                      | 9045        |
| 14                                 | EASTERN                            | 10303       |
| 15                                 | KOWLOON CITY                       | 12581       |
| 16                                 | TSUEN WAN                          | 12624       |
| 17                                 | YUEN LONG                          | 15439       |
| 18                                 | KWUN TONG                          | 17515       |

- Set the location of a certain station in a district, such as MTR station, and find out the two nearest specific kinds of indoor sports facilities.

```
1 select address, fac_name, ST_Distance(ST_SetSRID(ST_MakePoint(114.1, 22.4), 4326),point(lat, lon)::geography) as distance
2 From badminton_courts
3 Where fac_type LIKE 'Sports Centres'
4 and district LIKE 'TAI PO'
5 order by ST_Distance(ST_SetSRID(ST_MakePoint(114, 22), 4326),
6 point(lat, lon))
7 limit 2;
```

|   | address<br>character varying (256) 🔒                         | fac_name<br>character varying (256) 🔒 | distance<br>double precision 🔒 |
|---|--------------------------------------------------------------|---------------------------------------|--------------------------------|
| 1 | 6/F, Tai Po Complex, 8 Heung Sze Wui Street, Tai Po          | Tai Po Hui Sports Centre              | 8545.04324289                  |
| 2 | Podium Level, Tai Wo Shopping Centre, Tai Wo Estate, Tai ... | Tai Wo Sports Centre                  | 8414.05820007                  |



# Q&A session

**Thank You!**